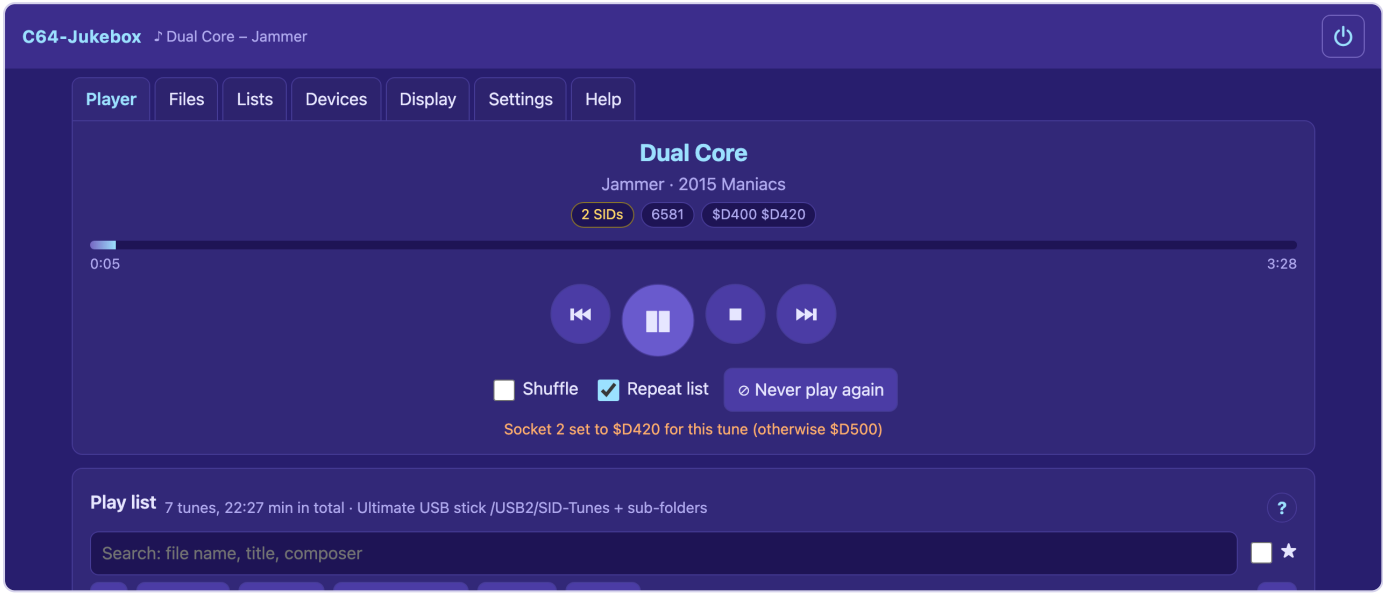
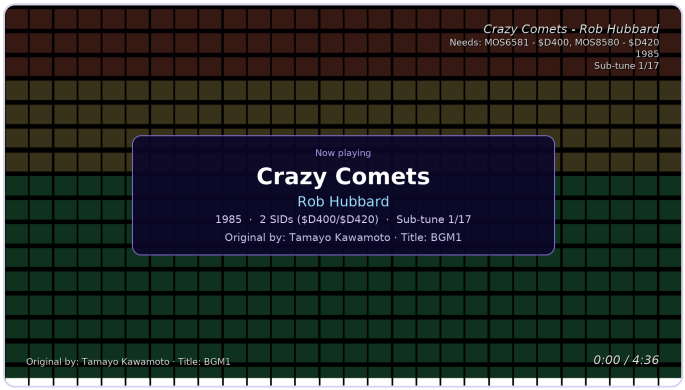
C64 Jukebox – Operation

Operating the SID jukebox on the Raspberry Pi: from the web page (phone or PC), on the TV with its remote control and on the touch display



The web page on the PC: current tune and play list



On the TV: an effect with the “Now playing” card



The menu on the TV, worked with the TV's remote

1 · Three ways to operate it
2 · The web page
3 · Choosing music
4 · Playback and the list
5 · Lists
6 · Devices
7 · Display: TV and screen
8 · Settings
9 · On the TV
10 · Remote control and menu
11 · Touch display on the Pi
12 · Switching on and off
13 · Quick help

This guide shows how to operate the jukebox (version 1.22). Installation, hardware and all details are in its manual – in the web page under the tab **Help** (every round **?** button jumps to the right place) and as `docs/MANUAL.md`. The pictures show sample data.

1 · Three ways to operate it

With	What you can do	Requirement
Web page phone, tablet, PC	**Everything:** choose and play music, lists, settings, TV, Ultimates, Picos, on/off.	browser in the same network, address **http://c64pi.local/**
The TV's remote control	Play, stop, change tune and sub-tune, change the effect – and through the **menu on the TV** choose music and search the HVSC, shuffle, picture and sound, Ultimate, update, language, switch off.	Pi on an HDMI input, HDMI-CEC switched on at the TV
Touch display on the Pi 2.8″, optional	Shows tune, levels and network; buttons ⏮ ■ ⏭ and ⏻ (TV on/off, switch off the jukebox).	display connected

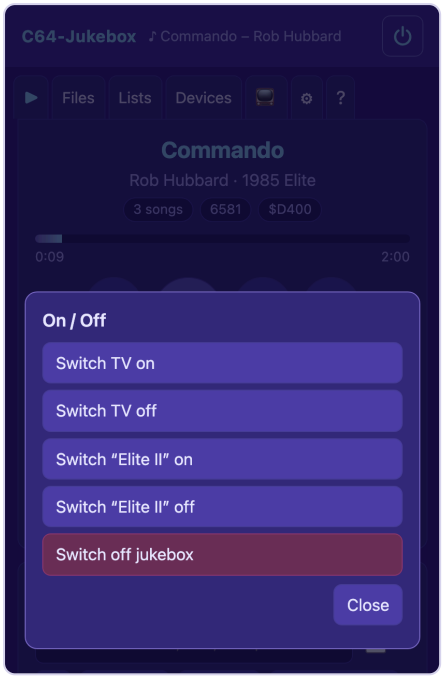
How to get to the web page:

1. Open **http://c64pi.local/** in the browser. If the name does not work (on some Android phones), use the Pi's IP address – it is shown on the TV next to the **QR code** (just scan it with the phone's camera) and on the Pi's display.
2. Log in with the web password (factory setting `commodore64` – a yellow bar reminds you to change it under *Settings → Web password*).
3. **As an app on the phone:** on an iPhone in Safari *Share → Add to Home Screen*, on an Android phone in Chrome ⋮ *→ Add to Home screen*. The jukebox gets an icon of its own (the note) and opens without an address bar like an app – no app store needed.

The jukebox keeps playing on its own, even when no page is open. All open pages show the same state – start on the phone and carry on at the PC.

English or German: on the first visit the jukebox takes the browser's language. To change it, use *Settings → System → Sprache / Language* or **Sprache / Language** in the TV menu; the language applies to the page, the TV menu, the displays and the manual.

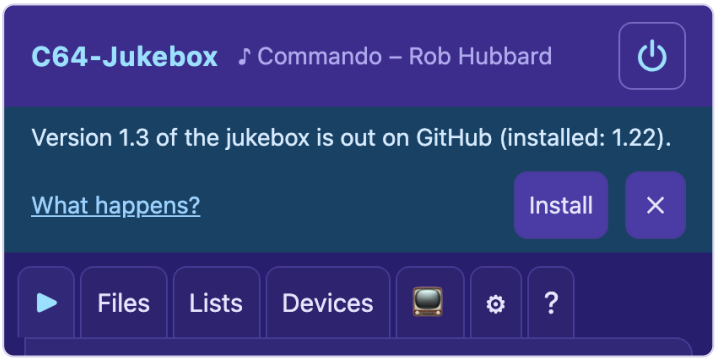
2 · The web page



The ⏻ button at the top right opens “On / Off” – from every tab

At the very top is always the **current tune**, with the ⏻ button to the right of it: **Switch TV on** / **off** (over HDMI-CEC), switch the selected **Ultimate** on and off, and **Switch off jukebox** (see section 12). Below are the tabs – on the phone partly as symbols:

Tab	Phone	Content
Player	▶	current tune, buttons, sub-tune, shuffle, play list
Files	Files	sources (NAS, FTP, USB sticks), folders, filter, search (also in the whole HVSC)
Lists	Lists	saved play lists, built lists, “Recently played”, statistics, blocked tunes
Devices	Devices	Ultimates (which one plays, on/off), Picos and stream distributor, devices on the network
Display	🖵	*Picture on the TV* · *Sound* (also Bluetooth headphones) · *Controls* (TV remote, display on the Pi)
Settings	⚙	*Playback* · *The Pi's network* · *System* (language, password, switching off, backup, updates)
Help	?	the manual

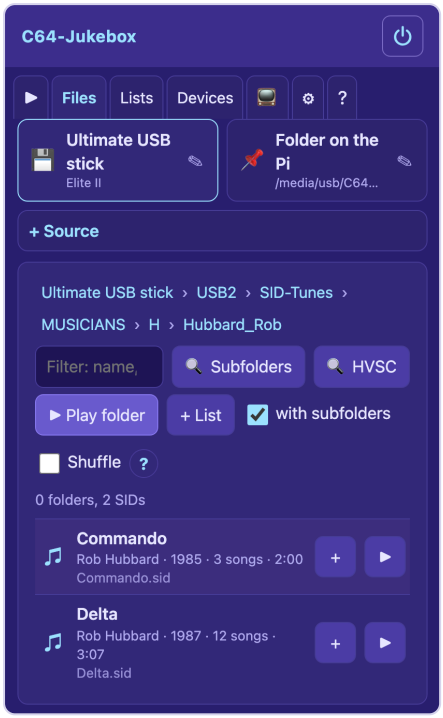


A new version on GitHub: a blue notice with **Install** (✕ hides it for this version)

Updates: once a day the jukebox looks for a new version on GitHub. It installs only when you say so – with **Install** in the notice, under *Settings* → *System* → *Updates* or in the TV menu (**Check for update**). This takes a few minutes; settings, lists and song lengths are kept, and if something goes wrong the old program comes back.

Short messages (“Saved”, “Commando – playing”) appear at the bottom for a few seconds.

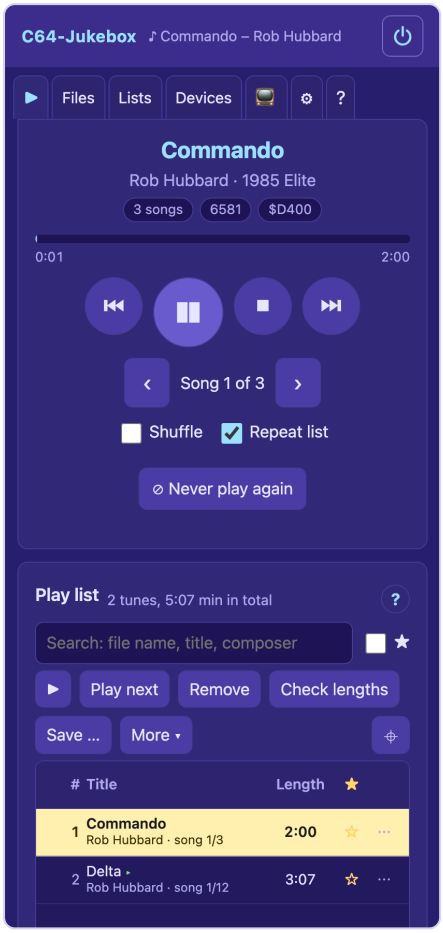
3 · Choosing music – tab “Files”



Sources at the top, below the open folder with title, composer, year and sub-tunes

- **Sources** are tiles at the top: a network share (NAS, Windows, Mac), an FTP server, the USB stick on the Ultimate or a USB stick on the Pi. Tap to open, ✎ edits, **+ Source** sets up a new one (with “Search the network”). The jukebox remembers the last folder of each source.
- Tap a **folder** to open it; the path line above leads back with one tap.
- **Tap a SID** (or ▶) to play it – and after it the other files of the folder. The current list is replaced.
- **+** appends a file to the list without interrupting playback.
- ▶ **Play folder** takes all SIDs of the folder – **with subfolders** from all subfolders too, with **Shuffle** mixed. **+ List** appends them instead.
- **Filter field:** works on the open folder at once (file name, title, composer, year; several words must all match, e.g. `hubbard 1986`).
- 🔍 **Subfolders** searches the word in **all subfolders** (up to 500 hits); “Play folder” then plays the hits, ✕ in the path line leads back.
- 🔍 **HVSC** searches the **whole HVSC** at once, without reading folders (as soon as the HVSC song lengths are loaded). It finds title and composer from the files once *Settings → Song lengths → Read titles for the search* has run.
- Yellow **2 SIDs** / **3 SIDs** marks stereo tunes.
- **Song lengths and STIL:** in the HVSC folder `C64Music/DOCUMENTS` tap **load as HVSC lengths** at `Songlengths.md5` and **load as STIL** at `STIL.txt`. The jukebox then changes exactly at the end of a tune and shows background notes on the tunes.

4 · Playback and the list – tab ▶

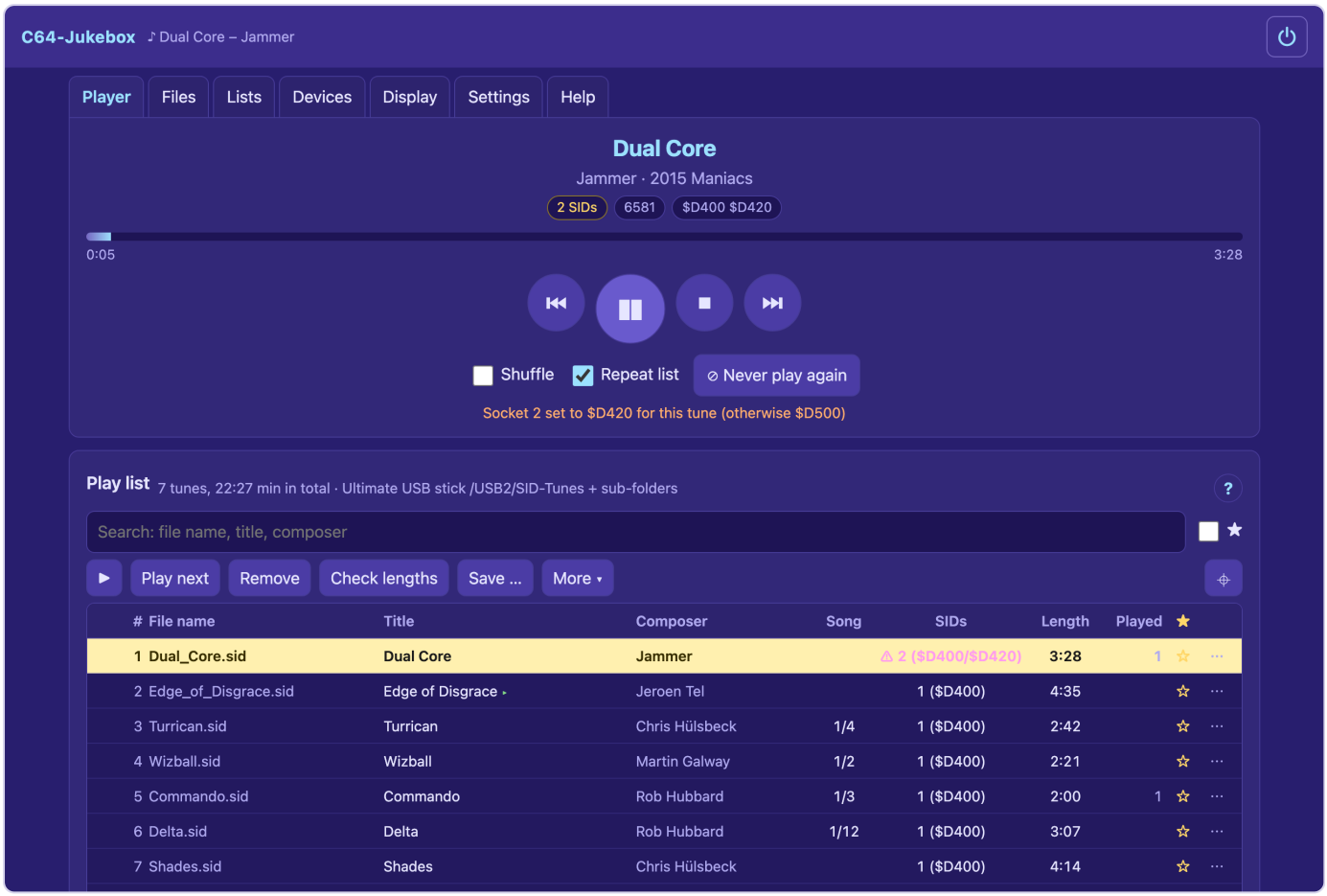


Current tune, buttons and the play list below (on the phone)

Button	Effect
⏮	to the start of the tune; within the first 5 s to the previous tune
▶ / ⏸	play or pause (the Ultimate is paused) and resume
■	stop – the Ultimate is reset
⏭	next tune
‹ Song 1 of 3 ›	another sub-tune of the current tune
Shuffle	mixes the order; tunes played less often come up sooner
Repeat list	start again at the end, otherwise the jukebox stops
Ultimate	with several Ultimates: which one plays

- **Progress bar:** with a known length to the end, otherwise to the time limit (“max. 5:00”).
- ⚑ **Mark end here** appears while a tune without a known length plays: press it at the end of the tune – the playing time becomes its length, and from then on it plays exactly that long.
- ⊘ **Never play again** skips to the next tune and puts the current one on the block list (see section 5).
- **Suggestions:** if the jukebox finds out by itself where a tune starts again, or finds the same tune in the HVSC, it offers to use that length (accept or **No**).
- Below the title: the SID chips, the STIL notes and, if needed, a SID socket change (“socket 2 at $D420 for this tune”).

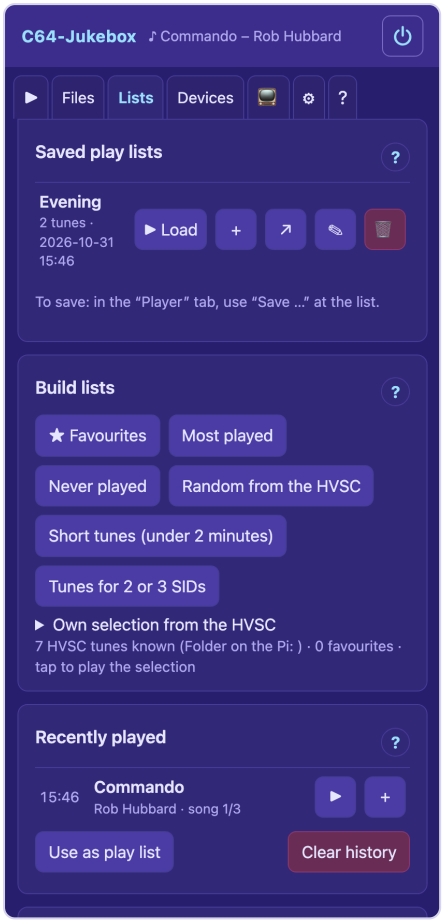
The play list



On the PC the list shows all columns: file name, title, composer, sub-tune, SIDs, length, times played, favourite. The current tune is yellow, a green ▸ marks the next one.

- **Play:** double-tap or double-click (on the PC also Enter).
- **Select:** tap; on the PC with Ctrl/⌘ several, with Shift a range.
- ⋯ at a tune (or right-click): play, play next, choose sub-tune, own length, favourite, never play again, show in the Files tab, remove from the list.
- **Play next** puts the selected tunes right after the current one; **Remove** takes them out.
- Tap ★ to make a tune a favourite; the ★ tick above the list shows favourites only.
- **Sort:** tap a column header; on the PC drag rows with the mouse.
- **Search:** the field above the list searches file name, title and composer.
- **Lengths** in the column: `3:05` from the HVSC, `3:05*` own, `(5:00)` grey = unknown (the time limit applies). **Check lengths** looks for missing lengths in the HVSC.
- **Save …** stores the list under a name (see tab “Lists”).
- **More ▾:** import/export a play list of the PC jukebox, **Discover** (a rarely played tune), **Stop after …** (tunes or minutes), select all, clear the list.
- **Keys on the PC:** Space play/pause, ← → previous/next tune.

5 · Lists – tab “Lists”



Saved play lists, built lists and “Recently played”

Saved play lists

▶ **Load** replaces the current list and plays it, **+** appends it, ↗ exports it for the PC jukebox, ✎ renames, 🗑 deletes. Lists are saved in the tab ▶ with **Save …**.

Build lists

One tap plays at once: ★ **Favourites**, **Most played**, **Never played**, **Random from the HVSC**, **Short tunes**, **Tunes for 2 or 3 SIDs**. Under **Own selection from the HVSC** you can choose by area (musicians, games, demos), composer, length, number of SIDs, STIL notes and “never played”. This needs the HVSC song lengths loaded from a source.

Recently played

The last 100 tunes with the time, newest first. ▶ plays one next, **+** appends it, **Use as play list** turns the history into a new list, **Clear history** empties it.

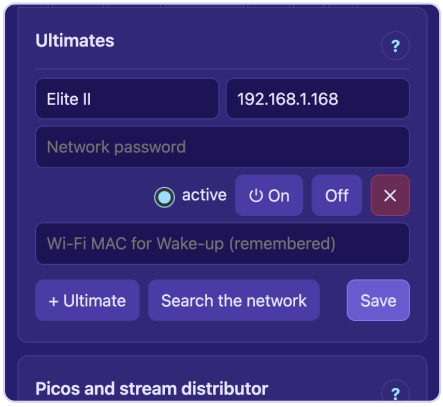
Statistics

How often what was played – counts and favourites belong to the file, also in other lists.

Blocked tunes

Tunes blocked with ⊘ **Never play again** (tab ▶, the ⋯ menu of a row or the TV menu). The jukebox skips them when moving on, with shuffle, “Discover” and in the built lists; in the play list they are crossed out. Chosen by hand they play anyway. **Play again** lifts the block.

6 · Devices – tab “Devices”

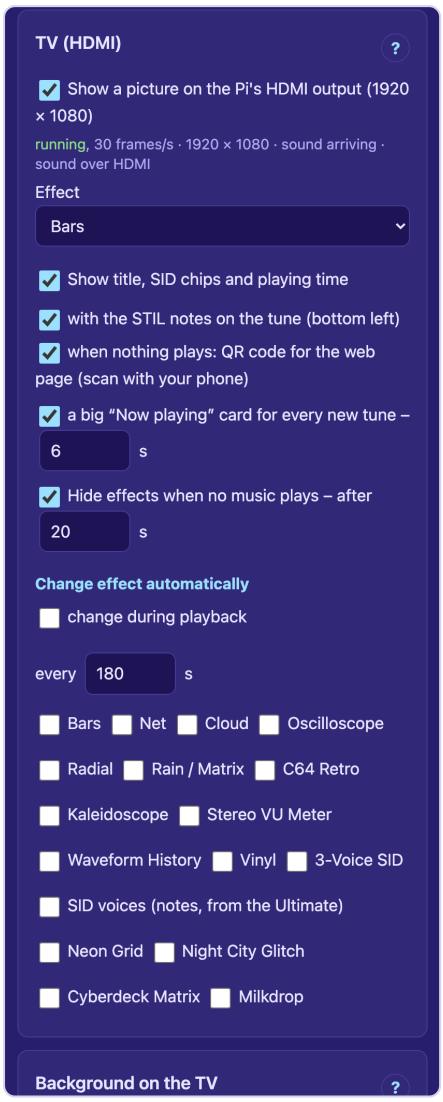


Card “Ultimates”

- **Ultimates:** name and **Ethernet address** of each Ultimate (**Search the network** finds them), the network password if set. **active** = this one plays. Don't forget **Save**.
- ⏻ **On / Off: Off** stops playback and switches the Ultimate off, **On** wakes it over its Wi-Fi (Ultimate 64-II). The jukebox learns the Wi-Fi MAC for this itself. The ⏻ button at the top of every page does the same.
- **Picos and stream distributor:** the Pi passes the Ultimate's sound on to visualizers and audio receivers and sends them titles and retro news – setup see the guide to the C64 ecosystem.
- **Devices on the network:** all Picos and the Pi with the way of the sound; **add as Pico** adds a device to the distributor's list.

7 · Display: TV and screen – tab 📺

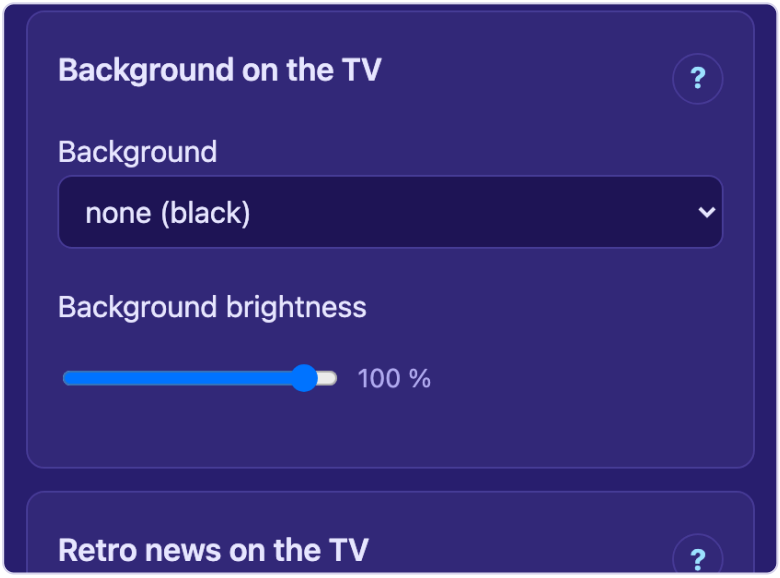
The tab has three groups: **Picture on the TV** (HDMI, background, retro news), **Sound** (sound on the TV, Bluetooth headphones) and **Controls** (TV remote, display on the Pi).



Card "TV (HDMI)"

Picture on the TV

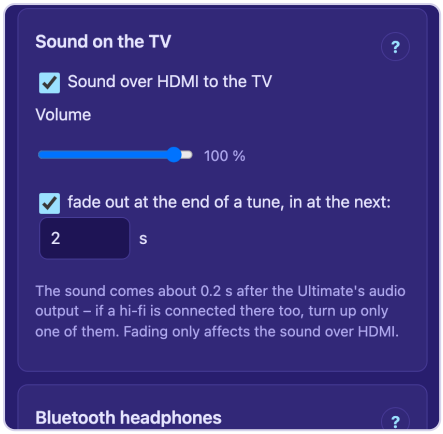
- **Show a picture on the Pi's HDMI output** switches the output on. Every change applies at once – you see on the TV right away how an effect looks.
- **Effect**: one of 17 (as in the PC jukebox) or "background only".
- **Overlays:** title, SID chips and playing time; STIL notes at the bottom left; a QR code when nothing plays; the big **"Now playing" card** for every new tune (duration adjustable).
- **Hide effects** when no music plays (after 20 s).
- **Change effect automatically:** tick effects and **change during playback** – a different one every 180 s.



Card "Background on the TV"

Background: none, **pictures** (slide show: upload them or choose a folder of a source, e.g. on the NAS; change every 20 s with cross-fade) or the **C64 picture** live from the Ultimate. "Background brightness" dims it.

Retro news on the TV: headlines from retro pages (Indie Retro News, amiga-news.de, Commodore News …) **when idle** as a big card, **as a ticker** at the bottom during the music. Every source can be switched on and off, own feeds with **+ Source**.



Card “Sound on the TV”

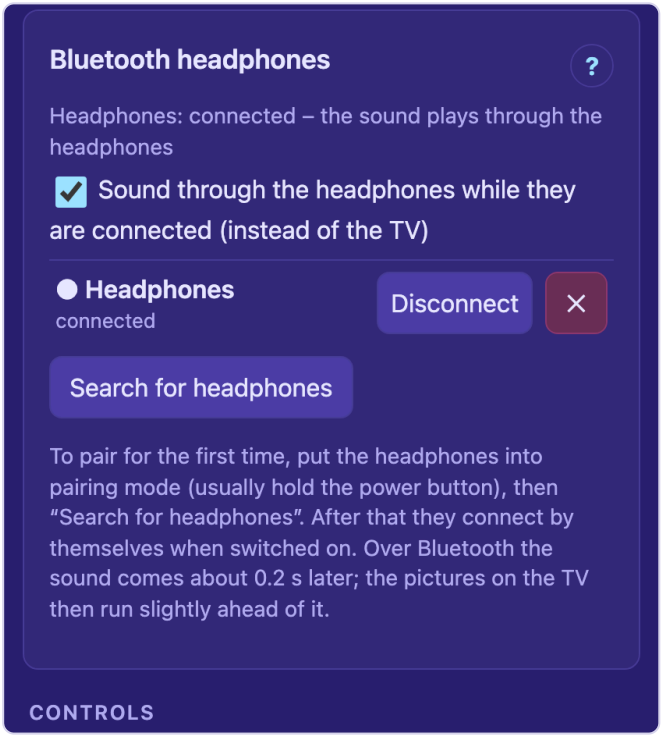
Sound on the TV

Sound over HDMI to the TV plays the Ultimate's sound through the TV. **Volume** (0–100 %) makes it quieter without looking for the remote. With **fade out at the end of a tune, in at the next** the tunes blend gently (only on the TV; the Ultimate itself plays unchanged). If a hi-fi is connected to the Ultimate as well, turn up only one of them – the TV is about 0.2 s later.

Bluetooth headphones

Instead of the TV, the sound can also play through Bluetooth headphones (or a Bluetooth speaker):

1. Put the headphones into **pairing mode** (usually hold the power button for a few seconds).
2. Tap **Search for headphones**, then **Pair** at the device found.
3. From then on the sound goes to the headphones as soon as they are connected, and back to the TV when you switch them off. Headphones that are switched on reconnect by themselves.



Card “Bluetooth headphones” with paired, connected headphones

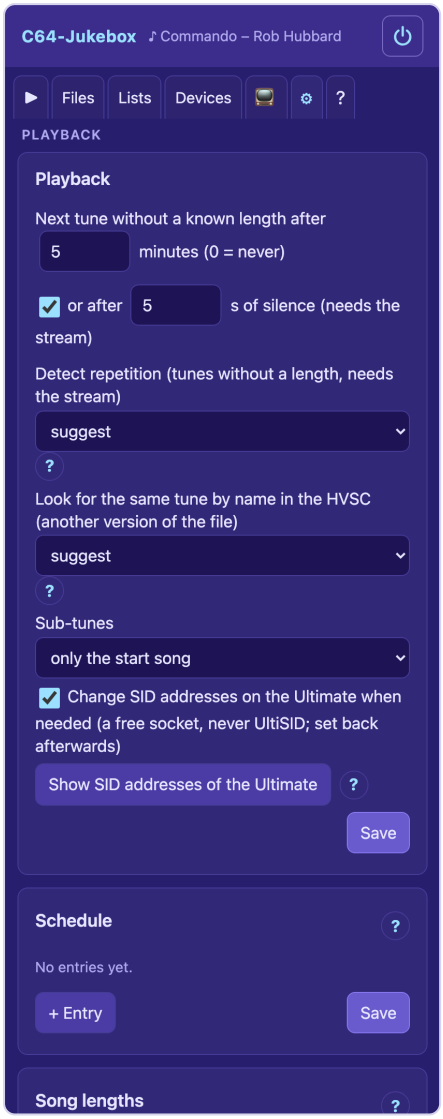
The HDMI output must be switched on for this. Volume and fading apply to the headphones too; on the TV, **Picture and sound** → **Sound via headphones** switches over.

Controls

TV remote control (HDMI-CEC): operate the jukebox with the TV's remote – see section 10. Also when the TV switches on and off: when the music starts, when the Pi starts, after a while without music – and whether the jukebox switches off when the TV goes off (see section 12).

Display on the Pi: type of display (small OLED or 2.8″ with touch), set up touch, rotate the picture, after how many minutes it goes dark – see section 11.

8 · Settings – tab ⚙



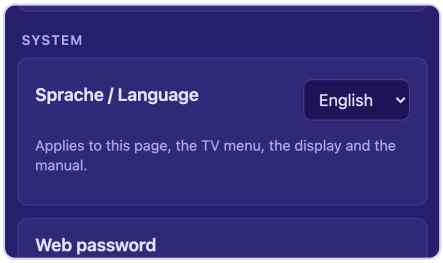
Playback: settings and schedule

Three groups: **Playback** (playback, schedule, song lengths, data from the PC jukebox), **The Pi's network** (Wi-Fi, party mode) and **System** (language, web password, switching off and smart plug, backup, updates).

Playback

Setting	Effect
Next tune without a known length after … minutes	time limit (default 5, 0 = never)
or after … s of silence	moves on when a tune ends and goes silent (needs the stream)
Detect repetition	suggests a length when a tune starts again – *suggest*, *automatic* or *off*
Look for the same tune by name in the HVSC	use the length of another version of the same music
Sub-tunes	*only the start song* or *all, one after the other*
Change SID addresses when needed	sets a free SID socket of the Ultimate to the address a stereo tune needs, and back afterwards

- **Schedule:** play a list or stop at fixed times, e.g. weekdays at 18:00 “Evening” – with the TV on/off if wanted.
- **Song lengths:** upload or remove the HVSC file, download own lengths, **Read titles for the search** (for 🔍 HVSC); remove STIL.
- **Take over data from the PC jukebox:** play counts, favourites and own lengths.
- **Wi-Fi and network:** choose a Wi-Fi (for the Pi without a cable), the Pi's own Wi-Fi; **party mode** for away from home without a router.



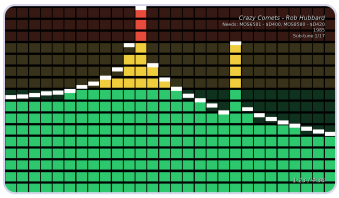
Card “Sprache / Language”

System

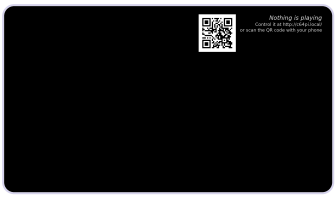
- **Sprache / Language:** English or German – for the page, the TV menu, the displays and the manual.
- Change the **web password**; download and restore a **backup** (all settings, sources, lists, own lengths and the block list in one file).
- **Switch off and smart plug:** switch off the jukebox, set up a smart plug – see section 12.
- **Updates:** look once a day (on/off), **Check now**, **Install**.

9 · On the TV

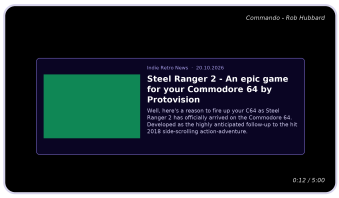
On the HDMI output the Pi shows effects to the music at 1920 × 1080 – over black, a slide show or the C64's live picture. **Top right** are title, composer, the SID chips needed, year and sub-tune, **bottom right** the playing time. For every new tune the **“Now playing” card** appears for a moment.



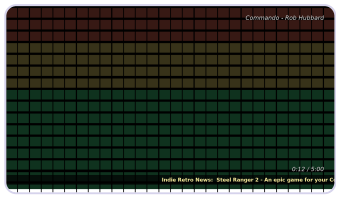
Effect “Bars” with the details top right



Nothing playing: note with a QR code to the web page

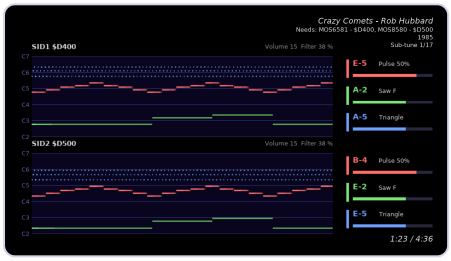


Retro news as a card when idle

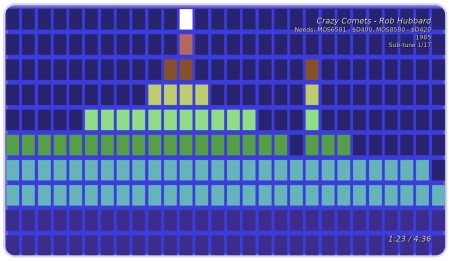


Retro news as a ticker during the music

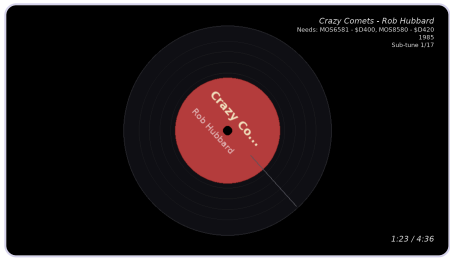
Some of the 17 effects



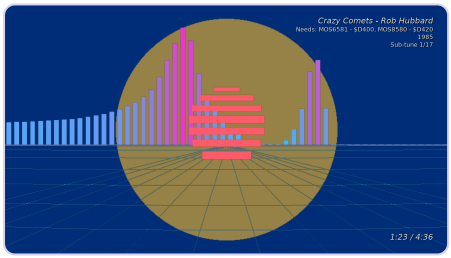
SID voices: the real voices of each SID as a piano roll



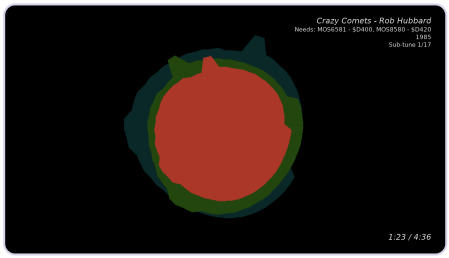
C64 Retro



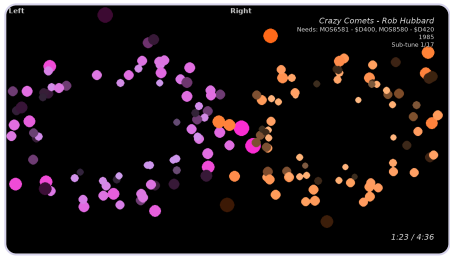
Vinyl



Neon Grid



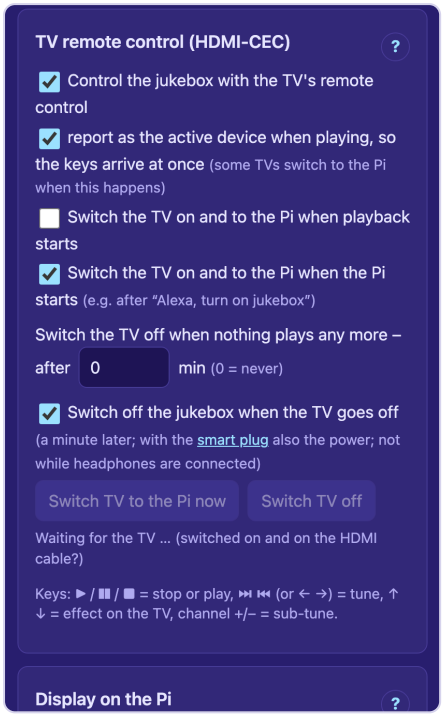
Milkdrop



Cloud

> **SID voices** shows the three voices of each SID for the last 5 seconds (pitch upwards, voice 1 red, 2 green, 3 blue, brighter = louder), on the right note, waveform and level. For this the Ultimate's **“Debug Stream Mode”** (under *Data Streams*) must include a 6510 part (default “6510 Only”), and it does not work together with the background “C64 picture”.

10 · Remote control and menu on the TV



Card "TV remote control (HDMI-CEC)" – also when the TV switches on and off

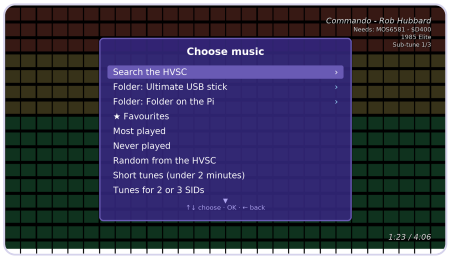
Switch on: tab 📺, card **TV remote control (HDMI-CEC)**, tick "Control the jukebox with the TV's remote control". CEC must be on at the TV (called e.g. *Anynet+*, *SimpLink*, *Bravia Sync*, *VIERA Link*). Most TVs send the keys only to the device they are showing – so switch to the Pi's input (or use **Switch TV to the Pi now** in the card).

Key	Effect
▶ ▮▮	stop while music plays – play when stopped
■	stop
⏭ ⏮ · ⏩ ⏪ · → ←	next / previous tune
↑ ↓	next / previous effect on the TV
Channel + –	next / previous sub-tune
OK or a menu key	opens the **menu on the TV**

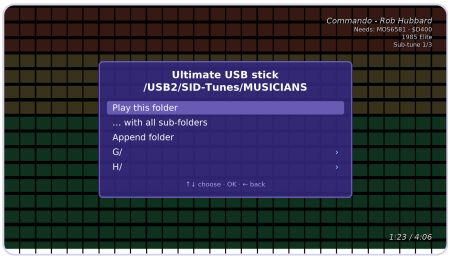
In the menu: ↑ ↓ choose, **OK** or → run or open, **Back** or ← one level back (at the top: close). After 30 s without a key it closes by itself. ▶, ■, ⏭ and ⏮ also work with the menu open.



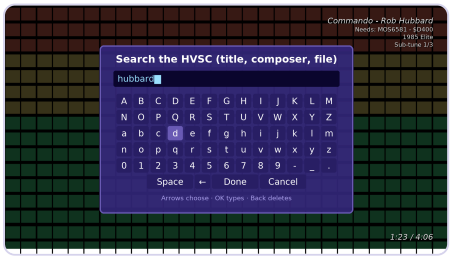
The main menu



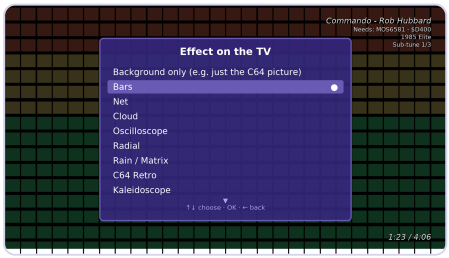
Choose music: HVSC search, sources, built lists, saved lists



Browsing the folders of a source



Searching the HVSC with the on-screen keyboard: arrows choose, OK types, Back deletes, "Done" searches



Choosing the effect (● = the current one)

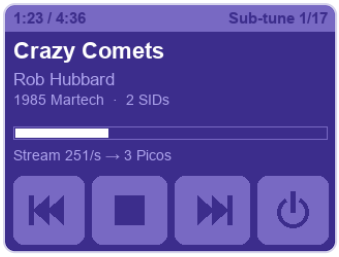


Switch off: TV, jukebox, Ultimate

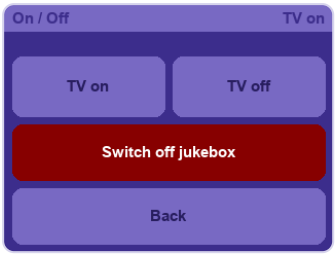
Menu item	What it does
Choose music ›	at the very top *Search the HVSC* (type words, “Done” shows up to 200 hits, *Play all …*), then the sources (“Folder: …”) to browse – *Play this folder*, *… with all sub-folders*, *Append folder*, a file plays the folder from there –, the built lists and all saved lists (play, append, rename, delete)
Mark end here / Use detected length / Use HVSC length	only for a tune without a known length: the playing time or the length found becomes its length
Never play again	asks first, puts the tune on the block list, then the next one comes
Shuffle, Repeat	on/off
Effect ›	choose the effect (switches the automatic change off)
Picture and sound ›	change effects automatically, show title, sound on the TV, sound via headphones (with paired headphones), volume on the TV (every OK one step quieter: 100 → 90 … 10 → 100)
Ultimate ›	which one plays (● = active), *Switch on* / *Switch off*, *Search the network*, add new ones with “+ name”
Check lengths	looks for missing lengths in the HVSC; then “Use … lengths”
Check for update	asks GitHub for a new version; if there is one, the line reads *Install update …* (asks first, the jukebox restarts)
Sprache / Language	switches between German and English
Switch off ›	*Switch TV off* (at once), *Switch off jukebox* and *Switch off Ultimate* (both ask first) – right after opening the menu reached with ↑ twice
Close	closes the menu

Notes appear discreetly above the playing time, e.g. “Starts again after 1:35 · OK: take as length” – **OK** then opens the menu at the right item, a second OK accepts.

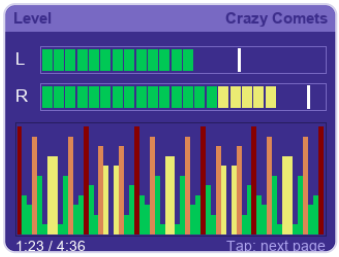
11 · Touch display on the Pi



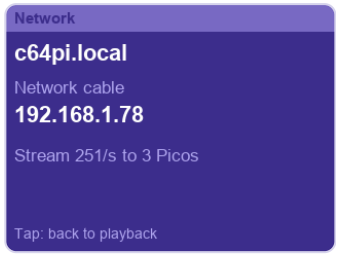
Playback with the buttons ⏮ ■ ⏭ ⏻



Button ⏻: the page “On / Off”



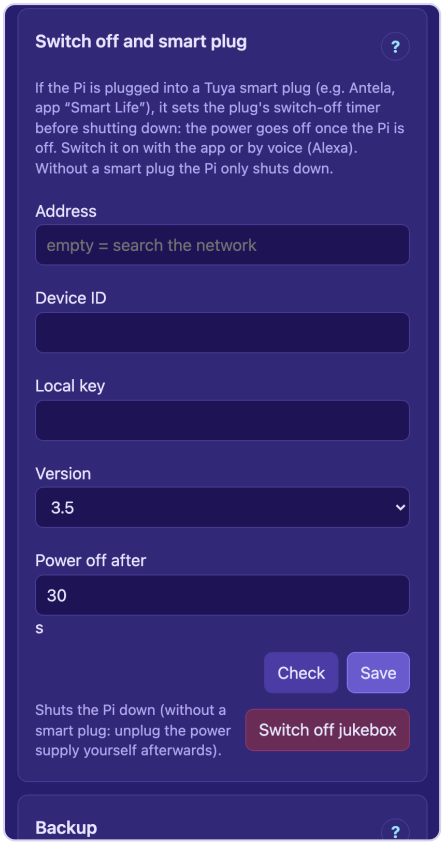
Levels left/right, below the last 15 s



Network: address, Wi-Fi, distributor, Ultimate

- **Buttons:** ⏮ previous tune, ■ stop (or ▶ play when nothing plays), ⏭ next tune, ⏻ on / off. A pressed button flashes briefly.
- **On / Off: TV on** and **TV off** (over HDMI-CEC; dark when the TV remote control is switched off), **Switch off jukebox** (red – only a second tap within 5 s switches off, the button then says “Tap again”) and **Back**. Top right shows whether the TV is on; after 30 s without a tap it returns to playback.
- **A tap outside the buttons** changes to the next page for 30 s: playback → levels → network.
- After 10 minutes without playback it goes dark (adjustable); **the first tap only wakes it**, the second one operates.
- The touch reacts to pressure – best tap with a fingernail or a stylus. **Set up touch** (tab 🖥️ , card “Display on the Pi”) shows three crosses to tap if taps land beside the buttons.
- In the Pi's own Wi-Fi the display shows name, password and address in large letters.

12 · Switching on and off



Card “Switch off and smart plug” (Settings → System)

Switching off the jukebox

Always shut the Pi down cleanly – simply cutting the power can damage the SD card. It works from everywhere:

Where	How
Web page	⏻ button at the top → **Switch off jukebox**, or *Settings* → *Switch off and smart plug*
TV	menu → **Switch off** → **Switch off jukebox** (asks first)
Touch display	⏻ → **Switch off jukebox**, tap twice

With a smart plug

If the Pi is plugged into a **Tuya smart plug** (app “Smart Life” or “Tuya Smart”), it sets the plug's switch-off timer before shutting down: after *Power off after* seconds (default 30) the power goes off. It is switched on with the app or by voice, e.g. “Alexa, turn on Jukebox” – the Pi starts as soon as there is power. To set it up, enter the plug's **Device ID** and **Local key** (where to find them is in the manual) and tap **Check**.

Together with the TV

In the card **TV remote control** (tab 📺):

- **Switch the TV on and to the Pi when the Pi starts:** the TV comes on and shows the Pi (in the first 90 s after the start).
- **Switch off the jukebox when the TV goes off:** one minute after the TV goes into standby the Pi shuts down, with a smart plug the power goes off afterwards – not while Bluetooth headphones are connected.
- Also: switch the TV on when playback starts; switch it off when nothing has played for a while.

So “Alexa, turn on Jukebox” is enough for the Pi and the TV, and in the evening the TV's remote switches everything off again.

Ultimate

Switch it on and off with the ⏻ button at the top, in the tab “Devices” or in the TV menu (*Ultimate* or *Switch off*).

13 · Quick help

What you see	What helps
`c64pi.local` does not work	Use the IP address: QR code on the TV or the Pi's display, otherwise in the router's list.
The remote does nothing	Switch the TV to the Pi's input (or “Switch TV to the Pi now”); switch CEC on at the TV; check the tick in the card “TV remote control”.
No sound on the TV, effects do not move	The sound comes as a stream from the Ultimate: is the Ultimate on and entered with its Ethernet address? Are *Data Streams* allowed in the Ultimate?
Headphones not found or silent	Keep the headphones in pairing mode while searching and pairing; the HDMI output must be switched on; otherwise tap **Connect**.
Tunes always run 5 minutes	Length unknown: load the HVSC song lengths (tab Files), press “Mark end here” at the end or accept the suggestions.
A voice is missing in a stereo tune	Switch on “Change SID addresses when needed”; ⚠ in the column “SIDs” marks tunes that do not match the Ultimate's setting.
SID voices: a note instead of notes	Switch on “Debug Stream Mode” at the Ultimate; not together with the background “C64 picture”.
A tune no longer comes up by itself	It may be on the block list: tab “Lists” → **Blocked tunes** → **Play again**.
Password forgotten	see the manual, chapter “Forgotten password” (reset over SSH).

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